CMPUT 195 Mini Project Report

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CMPUT 195: Introduction to the Principles and Techniques of Data Science

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1. Problem Formulation

This study examines the question: How does air pollution (PM2.5 exposure) affect life expectancy across countries?

The objective is to estimate whether higher pollution levels are associated with lower life expectancy, while accounting for key socioeconomic factors such as income (GDP per capita) and healthcare investment (health expenditure). This helps assess whether environmental conditions play a significant role in long-run health outcomes beyond traditional determinants.

Previous studies, such as that of Cao & Wu (2025), have proven that air pollution does have a negative association with life expectancy. However, these studies often focus on single country or region specific analyses.

A cross-country approach is important because the relationship between pollution and life expectancy may differ depending on economic development, healthcare access, and policy environments. By analyzing multiple countries simultaneously, this study can identify whether the negative effects of pollution are universal or moderated by national characteristics. This broader perspective improves the generalizability of findings and provides more relevant insights for global policy discussions on environmental health.

2. Data Acquisition

Data Source:

The data used in this study was obtained from the World Bank, which provides publicly available global development indicators; it is a widely used and reputable source for cross-country economic, environmental, and health statistics. We compiled four datasets which included annual observations for multiple countries containing key variables necessary to analyze the relationship between air pollution and life expectancy. The World Bank data is appropriate for this study because it ensures consistency in measurement across countries and provides sufficient coverage to support cross-country comparative analysis.

Data Description:

Our dataset is structured at the country-year level, meaning each row represents a specific country in a given year. This panel data structure allows for analyzing both cross-sectional differences between countries and variation over time.

The variables included in the dataset are:

- ***Country:*** The name of the country for each observation
- ***Year****:* The year in which the data was recorded.

- ***PM25****:* The average annual exposure to fine particulate matter (PM2.5), measured in micrograms per cubic meter (µg/m³). This is the key independent variable representing air pollution levels.
- ***LifeExp****:* Life expectancy (in years). This is the main dependent variable used to measure population health outcomes.
- ***GDP****:* Gross Domestic Product per capita (in current US dollars). This variable represents a country's economic development and income level.
- ***HealthExp****:* Health expenditure per capita (in current US dollars). This captures the level of investment in healthcare systems and services.

Relevance to the Research Question

This dataset is well-suited to the research question because it includes:

- A direct measure of air pollution (PM2.5)
- A clear health outcome (life expectancy)
- Key control variables (GDP and health expenditure) that account for socioeconomic differences across countries

By combining environmental, economic, and health indicators across multiple countries and years, the dataset provides sufficient variation and depth to estimate the relationship between pollution and life expectancy in a meaningful and relevant way.

3. Data Preprocessing

Initial Data Cleaning & Structuring:

Data was retrieved directly from the World Bank API and organized into a country-year panel dataset. Multiple datasets were merged using consistent country and year identifiers. Only variables relevant to the research question (PM2.5, life expectancy, GDP, and health expenditure) were retained.

Handling Missing Values:

Missing values were examined across variables and years. To improve data completeness and consistency the sample was restricted to the period of 2000-2020, where data availability is strongest. Observations with missing values for key variables (PM2.5, LifeExp, GDP, HealthExp) were also removed. This approach ensures that the final dataset contains complete cases for all variables used in the analysis, avoiding complications during modeling.

Duplicates and Data Consistency:

Duplicate country-year observations were identified and removed. Aggregate entities such as aggregates of countries with high or low incomes were excluded to avoid bias and ensure only individual countries were analyzed. The dataset was structured and sorted to form a clean panel structure, meaning one observation per country per year.

Categorical variables:

The Country variable is categorical and primarily used as an identifier rather than a predictor. Since the analysis focuses on regression with numerical variables, no explicit encoding (e.g., one-hot encoding) was required.

Numerical transformation and outliers:

To improve the quality of the analysis, log transformations were applied to PM2.5, GDP, and Health Expenditure, after ensuring that these variables are strictly positive. This transformation helps reduce skewness and limits the influence of outliers, it also allows coefficients to be interpreted in percentage terms.

Final Dataset and Validation:

The final dataset is a clean and consistent panel of country-year observations suitable for regression analysis.

Summary and Justification:

The preprocessing steps were designed to ensure data quality, consistency, and suitability for statistical modeling. Removing missing values and duplicates improves reliability, while restricting the time period increases data completeness. Log transformations help address skewness and outliers, making the relationships between variables easier to interpret. Overall, these steps ensure that the dataset is well-prepared for analyzing the relationship between air pollution and life expectancy.

4. Exploratory Data Analysis

Structure of the data

The dataset contains observations for 195 countries over the period 2000–2020, forming a country-year panel. It includes six main variables: Country, Year, PM2.5, Life Expectancy, GDP, and Health Expenditure. Data types and non-null counts were verified to confirm that the dataset is complete and properly formatted for analysis.

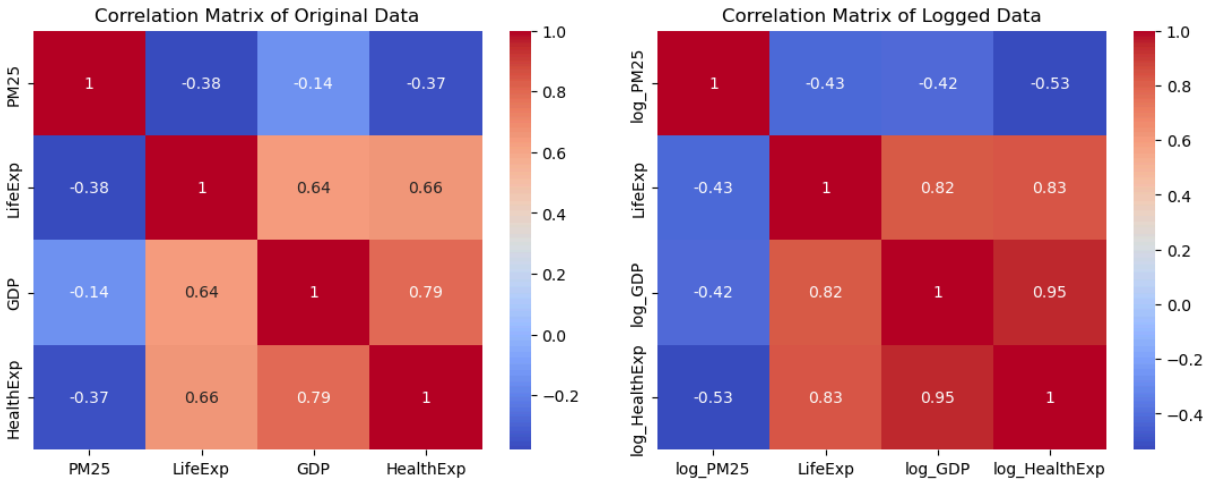
Statistical Analysis

Correlation matrices were computed for both raw and log-transformed variables to quantify relationships between key variables, as seen in Figure 1. The raw scatter matrix revealed

exponential relationships between GDP, health expenditure, PM2.5, and life expectancy, confirming that log transformations were appropriate. In the log-transformed data, life expectancy shows a strong positive correlation with log_GDP (r=0.82) and log_HealthExp (r = 0.83). A moderate negative relationship is observed between log_PM2.5 and life expectancy. log_GDP and log_HealthExp were found to be almost perfectly correlated (r = 0.95), indicating potential multicollinearity between the two control variables.

Figure 1.

Correlation matrices for Original (Raw) and Log-Transformed data



<u>Significance:</u>

These results suggest that while air pollution is related to life expectancy, economic factors such as income and healthcare spending are also very strong predictors. The high correlation between GDP and health expenditure highlights the need to carefully interpret regression results, as their effects may overlap.

<u>Additional Observations</u>

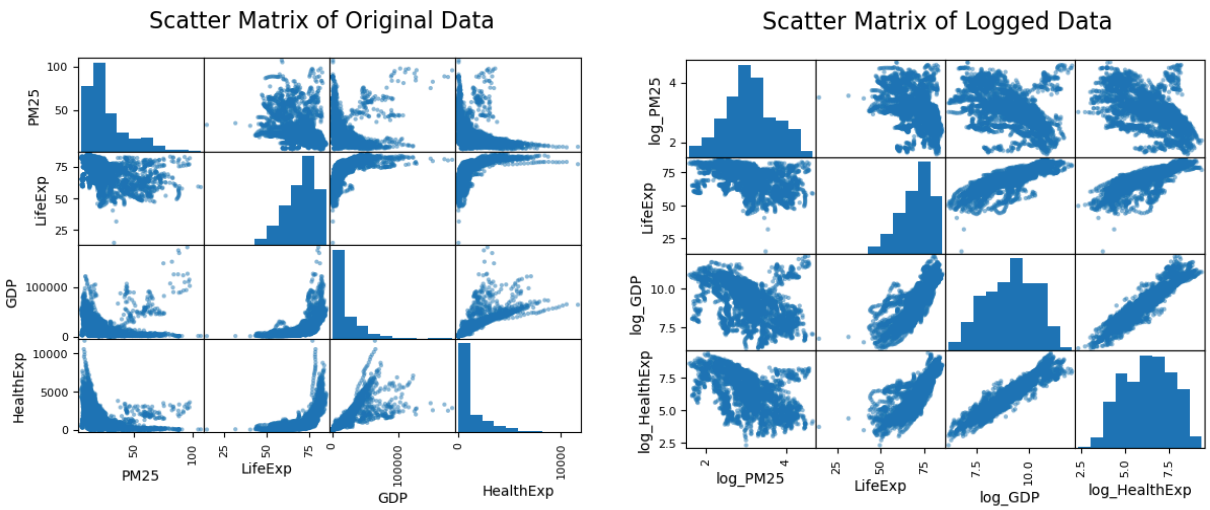
- Global average life expectancy increased steadily from around 67 years in 2000 to approximately 73 years in 2019, before a slight dip in 2020
- Average GDP and health expenditure increased consistently over the period, with health expenditure nearly tripling from 2000 to 2020
- Average PM2.5 has been trending downward, with a few spikes, indicating that overall air quality is improving

Visual analysis

<u>1. Scatterplots - raw and log-transformed</u>

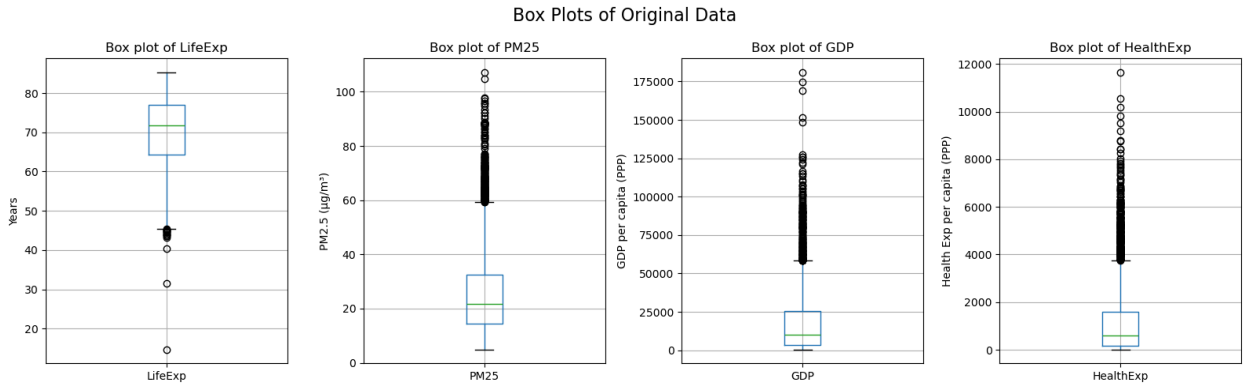
Scatter matrices, shown in Figure 2, were produced for both raw and log-transformed variables to identify the nature and direction of relationships between all variables. In the raw data, relationships appear non-linear (exponential), after log transformation, relationships become more linear and interpretable. This confirms that log transformations are appropriate and supports their use in regression modeling.

Figure 2.
Scatter matrices for Original (Raw) and Log-Transformed Data



<u>2. Boxplots (Distribution and Outliers)</u>
Boxplots of the original data were generated for PM2.5, Life Expectancy, GDP, and Health Expenditure, as seen in Figure 3. Life expectancy is centered around 70 years, with notable variation and a few low outliers. PM2.5, GDP, and health expenditure are all right-skewed, with the majority of countries clustered at lower values and a small number of high outliers. GDP and health expenditure show a wide spread at the high end, reflecting the large gap between low and high-income countries. Due to the skewness of PM2.5, the median is used as the measure of center for the hypothesis tests. The overall skewness justifies the use of log transformations and highlights inequality between countries, which is important when comparing global health outcomes.

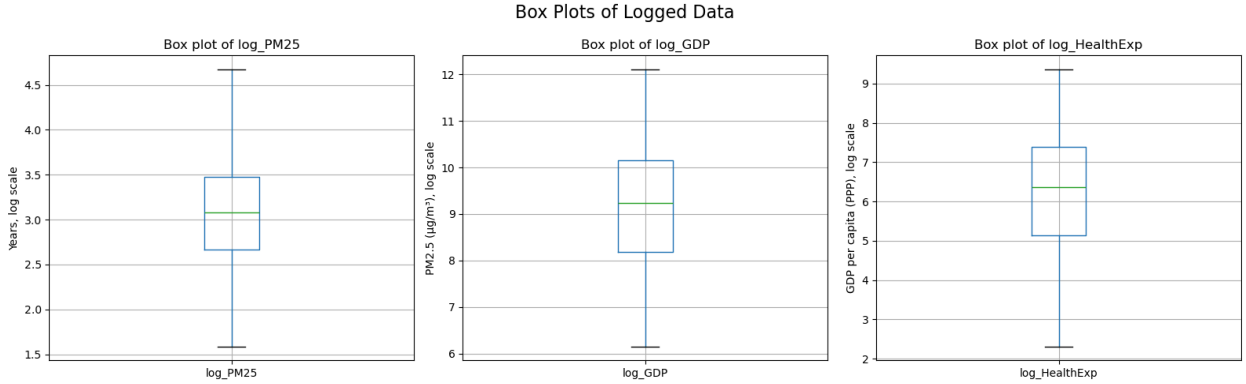
Figure 3.
Boxplots of all four variables of the Original data (LifeExp, PM25, GDP, HealthExp)



Boxplots were then generated for the three log transformed variables, these can be seen in Figure 4. These boxplots show how the log-transformation to these variables reduces skewness and outliers. However we continue to use the original variables for EDA as the relationships between variables do not change meaningfully, and the original values are easier to interpret.

Figure 4.

Boxplots of the three log-transformed variables (log_PM25, log_GDP, log_HealthExp)

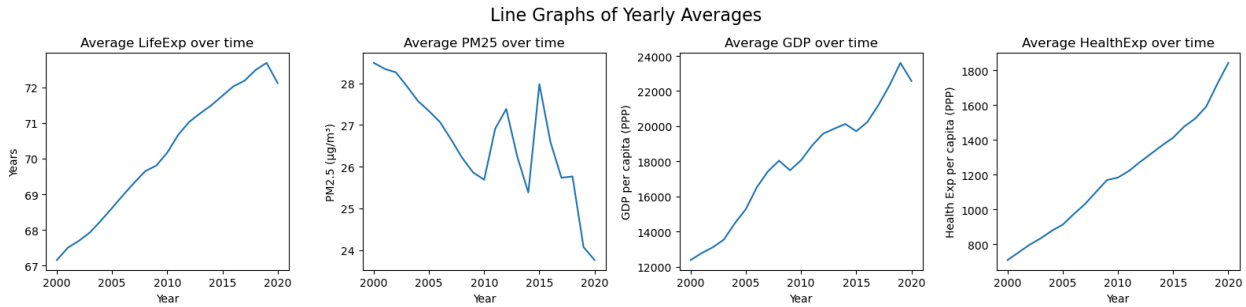


<u>3. Trends – Global Averages over time</u>

Figure 5 displays line graphs constructed to examine global average trends in PM2.5, life expectancy, GDP, and health expenditure from 2000 to 2020. We observed that global average life expectancy increased steadily from approximately 67 years in 2000 to around 73 years in 2019, followed by a slight decline in 2020. GDP similarly followed a steady increase from 2000 to 2019, with a slight dip in 2020. This slight decrease in 2020 for life expectancy and GDP could be attributed to the global Covid-19 pandemic, which could impact both those variables. Health expenditure shows a consistent upward trend, increasing substantially over time. PM2.5 levels show an overall downward trend over time, with some fluctuations.

Figure 5.

Global average trends of the four variables (LifeExp, PM25, GDP, HealthExp)



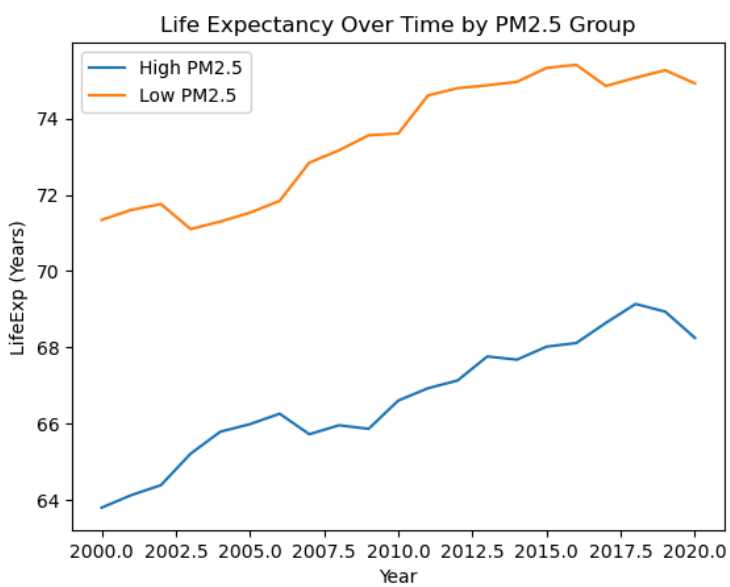
These trends suggest that global health outcomes have improved over time alongside economic growth and increased healthcare spending. At the same time, improving air quality may also contribute to rising life expectancy. However, because all variables change over time, this highlights the importance of using regression analysis to isolate the individual effect of pollution.

4. Life Expectancy by PM2.5 Group

Figure 6, a line graph comparing life expectancy over time for countries with high vs low PM2.5 exposure, with countries being separated based on whether they sit above or below the mean PM2.5 level, shows a consistent gap between the two groups. Countries with low PM2.5 consistently have higher life expectancy and the gap between the groups remains relatively stable over time, with no major reversals.

Figure 6.

Life expectancy over time by PM2.5 group



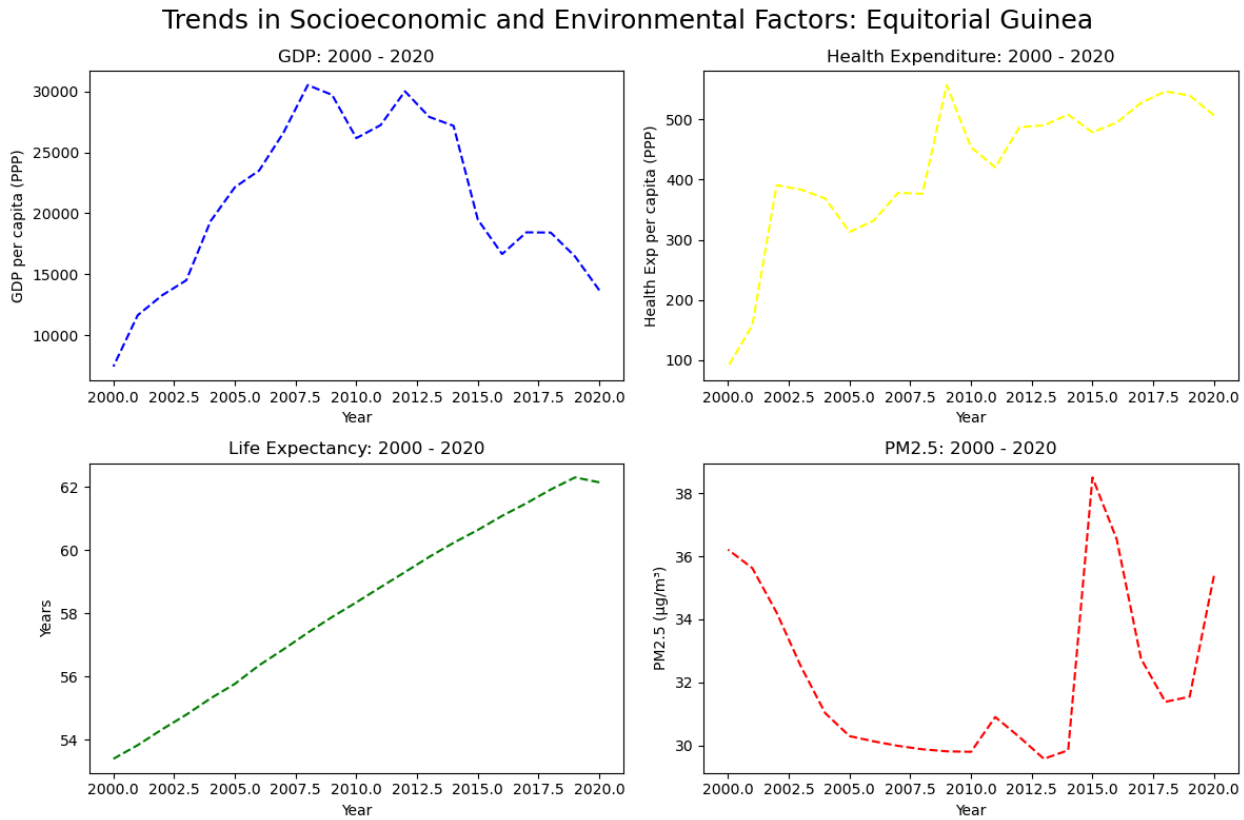
This provides strong visual evidence between air pollution and life expectancy. The persistence of the gap suggests that the relationship is not driven by short term fluctuations, but reflects a more stable, long term pattern.

3. Country Case Study: Equatorial Guinea

A 2x2 subplot, seen in Figure 7, examined trends in GDP, health expenditure, life expectancy, and PM2.5 over time for Equatorial Guinea. This revealed that GDP declined after 2015, health expenditure remained relatively low, while life expectancy increased gradually. This example shows that high GDP alone does not guarantee better health outcomes, reinforcing the importance of including multiple control variables.

Figure 7.

2x2 subplot of trends for the four variables in Equatorial Guinea over time (GDP. HealthExp, LifeExp, PM2.5)

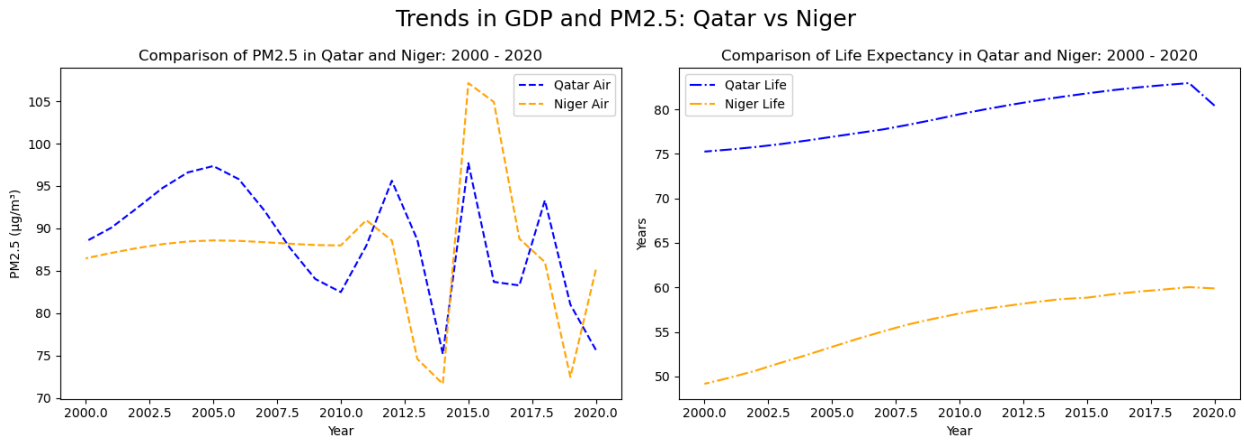


4. Country Comparison: Qatar vs Niger

Figure 8 shows side-by-side line plots comparing Qatar and Niger over time, which showed that, despite similar PM2.5 levels, Qatar consistently has higher life expectancy. This visually demonstrates that GDP likely confounds the relationship between pollution and life expectancy, supporting the need to control for economic factors in the model.

Figure 8.

Comparison of PM25 and Life Expectancy between Qatar and Niger over time



Hypothesis Testing

A permutation test was conducted to assess whether the difference in life expectancy between high and low PM2.5 countries was statistically significant.

- *Null hypothesis:* There is no difference in mean life expectancy between high and low PM2.5 countries
- *Alternative hypothesis:* Mean life expectancy is lower in high PM2.5 countries.

5,000 permutations were performed by shuffling PM2.5 group labels and recomputing the difference in means each time. The resulting p-value was 0.0, which is below the 0.05 significance level, meaning the null hypothesis is rejected. This means there is strong statistical evidence that countries with higher pollution tend to have lower life expectancy, which supports the main research question.

Relationship Insights

Among the 50 lowest-GDP countries, 40 also had the lowest life expectancy, the majority being African countries, with Afghanistan and Pakistan as exceptions. Conversely, 40 of the 50 highest-GDP countries had the highest life expectancy, with a concentration of European countries. Equatorial Guinea was the only exception in the high-GDP group; despite a high GDP, its life expectancy remains low, likely due to its disproportionately low health expenditure and relatively high PM2.5 levels. Many countries with both low life expectancy and high pollution also have low GDP. These patterns reinforce that GDP is a key confounding variable, and failing to control it would overstate the impact of pollution.

5. Modelling and Problem Formulation

Model Choice and Justification

We are using an Ordinary Least Squares (OLS) regression model to analyze the relationship between air pollution and life expectancy. OLS is appropriate for a few reasons. The first is that

the dependent variable, life expectancy, is continuous. Secondly, the goal is to estimate the magnitude and direction of relationships between variables. Lastly, insights from the EDA showed approximately linear relationships after log transformation, satisfying a key assumption of linear regression

Model Specification

1. Baseline Model

The baseline model examines the direct relationship between air pollution and life expectancy.

- Equation:
  - $LifeExp_i = \beta_0 + \beta_1 log_PM25_i + \epsilon_i$
- *Dependent Variable:* LifeExp
- *Independent Variable:* log_PM25

This model provides a simple estimate of how pollution affects life expectancy without accounting for other factors. It establishes a baseline relationship that can later be compared to more complete models.

2. Multivariate Model

The second model includes key socioeconomic controls.

- Equation:
  - $LifeExp_i = \beta_0 + \beta_1 log_PM25_i + \beta_2 log_GDP_i + \beta_3 log_HealthExp + \epsilon_i$
- *Dependent Variable:* LifeExp
- *Independent Variables*: log_PM25, log_GDP, log_HealthExp

Variable selection:

- log_PM25 (Air Pollution)
  - This is the main variable of interest. It captures exposure to air pollution and is expected to have a negative effect on life expectancy based on prior research and EDA findings.
- log_GDP (Income Level)
  - Included to control for differences in economic conditions across countries. EDA showed a strong positive relationship between GDP and life expectancy, meaning failing to include it could bias the estimated effect of pollution.
- log_HealthExp
  - This variable captures investment in healthcare systems. Countries with higher health expenditure tend to have better health outcomes, so it is important to control for this factor.

Consideration of multicollinearity:

- EDA revealed a very high correlation (r = 0.95) between log_GDP and log_HealthExp, indicating potential multicollinearity. While both variables are theoretically important, their high correlation may make it difficult to separately identify their individual effects. However, both are retained in the model because they represent conceptually different

factors (income vs healthcare investment), and excluding one may introduce omitted variable bias.

Summary

Two models are used. The first is a baseline model to measure the raw relationship between pollution and life expectancy, and the second is a multivariate model that controls for key socioeconomic variables. This approach allows for comparison between models and helps determine whether the effect of air pollution remains significant after accounting for economic and healthcare differences across countries.

6. Model Summary

Training Procedure

The dataset was split into training (80%) and testing (20%) subsets using a randomized split. The model was fit using only the training data to learn the relationships between variables, while the test set is reserved for evaluating model performance.

Baseline Model (OLS)

The baseline model can be defined mathematically as:

$$\widehat{LifeExp}_i = 89.506 - 6.255 \cdot log_PM25_i$$

<u>Interpretation of coefficients</u>

- The intercept (89.506) represents the life expectancy when pollution is at 1µg/m³ (log = 0)
- The coefficient on log_PM25 (-6.255) indicates that higher pollution is associated with lower life expectancy

Multivariate Model (OLS)

The baseline model can be defined mathematically as:

$$\widehat{LifeExp}_i = 26.742 - 0.370 \cdot log_PM25_i + 2.828 \cdot log_GDP_i + 2.971 \cdot log_HealthExp_i$$

<u>Interpretation of coefficients</u>

- log_PM25 (-0.370): After controlling for economic and healthcare factors, pollution still has a negative effect on life expectancy, though the magnitude is much smaller than in the baseline model
- log_GDP (2.828): Higher income levels are strongly associated with higher life expectancy
- log_HealthExp (2.971): Greater healthcare spending is also associated with longer life expectancy

Model Comparison

The baseline model has an R^2 of 0.183, which indicates limited explanatory power. While the multivariate model has an R^2 of 0.704, showing a substantial improvement after adding control

variables. This suggests that much of the variation in life expectancy is explained by economic and healthcare factors, and that failing to include them would overstate the effect of pollution.

7. Model Evaluation

Evaluation was performed on our second model, the multivariate model that controls for the socioeconomic factors.

Test Set and Generalization

The dataset was split into a training set (80%) and a test set (20%) using random sampling. The test set was not used during model training, allowing for an unbiased evaluation of how well the model generalizes to unseen data.

Performance Metrics

Two metrics were used to evaluate model performance.

1. Coefficient of Determination (R^2), which measures the proportion of variance in life expectancy that is explained by the model
2. Root mean square error (RMSE), which measures the average prediction error in units of life expectancy (years)

These metrics are appropriate because the problem is a regression task with a continuous outcome variable.

Model Performance

Training Set performance:

- Train R^2: 0.708
- Train RMSE: 4.854

Test Set Performance:

- Test R^2: 0.687
- Test RMSE: 4.994

Interpretation of metrics

<u>R^2 (Test = 0.687)</u>

The model explains approximately 68.7% of the variation in life expectancy across countries. This indicates a strong fit, especially given the complexity of health outcomes, which are influenced by numerous factors.

<u>RMSE (Test = 4.993)</u>

On average, the models predictions are off by around 5 years of life expectancy. This is a reasonable error given that the overall mean life expectancy is 70 years.

Training vs Testing Comparison
The training R^2 (0.708) and testing R^2 (0.687) are very close, the training RMSE (4.854) and testing RMSE (4.994) are also similar. This indicates that the model is not overfitting and generalizes well to unseen data. The small drop in performance from training to testing is expected and suggests the model is capturing real underlying patterns rather than noise.

Overall Evaluation
The model performs well in predicting life expectancy, with strong explanatory power and reasonable prediction error. While not perfect, it captures the majority of variation in the data and provides reliable estimates for analyzing the relationship between air pollution and life expectancy.

8. Interpretation of Results and Conclusion

Interpretation of Results
Overall, the model was successful in explaining variation in life expectancy across countries. The multivariate model achieved an R^2 of approximately 0.70, indicating that a large portion of the differences in life expectancy can be explained by air pollution, income, and healthcare spending.

The results show that PM2.5 exposure has a statistically significant negative relationship with life expectancy. In the baseline model, a 1% increase in PM2.5 is associated with a 6.255 year decrease in life expectancy. However, after controlling for economic and healthcare factors, this effect decreases to 0.370 years, suggesting that part of the initial relationship was driven by omitted variables such as income. This highlights an important insight that while pollution does negatively affect health outcomes, economic conditions and healthcare investment have a much larger impact. The coefficient on log_GDP (2.828) indicates that a 1% increase in GDP per capita is associated with an increase of approximately 2.83 years in life expectancy, and the coefficient on log_HealthExp (2.971) suggests a similarly strong positive relationship between healthcare spending and life expectancy.

These findings imply that wealthier countries with higher healthcare investment tend to achieve significantly better health outcomes, even when pollution levels are similar.

Key Insights

Our key insights are:
- Air pollution has a negative and statistically significant effect on life expectancy
- The effect of pollution becomes much smaller after controlling for GDP and healthcare, indicating confounding
- GDP and health expenditure are the strongest predictors of life expectancy in the model

- The model explains a large portion of variation across countries, suggesting good overall fit

Strength of the Model

A key strength of our model is its ability to incorporate multiple important factors simultaneously, allowing for a more accurate estimate of the effect of air pollution. Additionally, the model demonstrates strong predictive performance ($R^2 \approx 0.70$) and generalizes well to unseen data, as shown by similar training and testing results.

Limitation of the Model

One important limitation is the presence of multicollinearity between GDP and health expenditure, as identified in the EDA. This makes it difficult to clearly separate their individual effects on life expectancy.

Additionally, the model may suffer from omitted variable bias, as other relevant factors, such as education, diet, or access to clean water, are not included in the analysis.

Conclusion

This study finds that air pollution is an important determinant of life expectancy, but its impact is relatively small compared to broader socioeconomic factors. While reducing pollution can improve health outcomes, policies aimed at increasing income levels and healthcare access may have a larger overall effect on life expectancy.

(Bonus) Recommendations and Future Work

Future researchers could improve this analysis in a few ways. The first would be by including additional variables such as education levels, urbanization or access to clean water. They could also use panel data techniques like fixed effects models to better control for unobserved country specific differences. Finally, they could explore causal methods to better isolate the true effect of air pollution.

If this study were repeated, a more advanced modeling approach could be used to address multicollinearity and better capture country specific effects over time.

9. Documentation and Citations

Bibliography

Cao, Y., & Wu, J. (2025). Impacts of wildfire smoke PM2.5, greenspace and terrain ruggedness on life expectancy in the contiguous United States. *The Science of the Total Environment*, *964*, 178584. https://doi.org/10.1016/j.scitotenv.2025.178584